Air Quality Analysis Across Indian States: Pollutant Distribution, Socioeconomic Correlates, and Predictive Modelling

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***Abstract*—Air pollution is a critical public health and policy challenge in India, where rapid urbanisation and industrial growth have placed acute pressure on ambient air quality. This study analyses real-time pollutant concentration data from the Central Pollution Control Board (CPCB) monitoring network, integrating findings with state-level socioeconomic and health indicators drawn from the National Family Health Survey (NFHS-5, 2019–21). Seven major pollutants — PM_{10}, $PM_{2.5}$, CO, NO_2, NH_3, O_3, and SO_2 — were examined across 3,273 valid monitoring observations spanning 29 states and 267 cities. Key findings indicate that particulate matter dominates the pollution landscape, with mean PM_{10} and $PM_{2.5}$ concentrations of 96.11 µg/m^3 and 76.60 µg/m^3, respectively — exceeding WHO annual guideline values by six- and fifteenfold. Delhi, Himachal Pradesh, and Jharkhand recorded the highest composite pollution averages. Pearson correlation analysis with 29 matched state-level profiles reveals that infant and under-five mortality are the only indicators reaching statistical significance ($r = +0.45$, $p = 0.017$), while development indicators show weak inverse trends. K-Means clustering identified three distinct pollution-vulnerability state profiles. Predictive modelling via Pipeline-based cross-validation confirms that socioeconomic proxies alone are insufficient for reliable state-level pollution prediction at $n = 29$, pointing to the need for longitudinal city-level panel data. The study contributes an integrated, reproducible analytical framework applicable to future large-scale air quality research in low- and middle-income country contexts.**

***Index Terms*—Air Quality Index, $PM_{2.5}$, PM_{10}, India, NFHS-5, K-Means Clustering, Random Forest, Socioeconomic Indicators, CPCB, Predictive Modelling**

I. Introduction

India's air quality crisis has intensified over the past two decades, driven by a complex interplay of industrial emissions, vehicular exhaust, agricultural burning, and household fuel use [1]. The Global Burden of Disease study estimates that ambient and household air pollution together account for over 1.6 million deaths annually in India [2]. Despite significant regulatory effort — including the National Clean Air Programme (NCAP) launched in 2019 — comprehensive, state-level evidence on pollution distribution and its socioeconomic determinants remains fragmented.

Two primary data resources enable this analysis. The Central Pollution Control Board (CPCB) operates a real-time Continuous Ambient Air Quality Monitoring (CAAQM) network covering hundreds of monitoring stations across India. The National Family Health Survey (NFHS-5), conducted by the Ministry of Health and Family Welfare in 2019–21, provides a rich repository of household-level socioeconomic, health, and demographic indicators for all states and union territories.

Prior studies have examined individual pollutants [3], city-level AQI trends [4], or isolated health correlates [5], but few have systematically integrated CPCB real-time data with NFHS-5 socioeconomic profiles at the state level. This study addresses that gap with the following objectives:

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1) **Characterise** the distribution of seven key pollutants across Indian states and classify monitoring readings by standard CPCB AQI categories.
2) **Identify** socioeconomic and health indicators most strongly correlated with state-level pollution exposure.
3) **Cluster** states into distinct pollution-vulnerability profiles using K-Means clustering.
4) **Build and evaluate** predictive models for state-level mean pollution using socioeconomic features, identifying the principal determinants.

The remainder of the paper is structured as follows. Section II describes the datasets. Section III details the methodology. Section IV presents results and discussion. Section V concludes with limitations and future work.

II. Dataset Description

A. CPCB AQI Dataset (`aqi.csv`)

The AQI dataset was sourced from the Government of India's open data portal (data.gov.in) and represents real-time readings from CPCB's CAAQM network. It contains 3,411 observations across 11 variables. Table I summarises the key variables.

Coverage: 29 states, 267 cities, 7 pollutants. The dataset snapshot is dated 27 March 2026 (UTC+5:30). It contains 138 rows (≈4%) with `pollutant_avg` reported as `NA`, which were excluded, leaving 3,273 valid observations.

TABLE I
CPCB AQI DATASET VARIABLES

Variable	Description
`state`	State / union territory name
`city`	City of the monitoring station
`station`	Full station name (operator)
`last_update`	Timestamp (DD-MM-YYYY HH:MM:SS)
`latitude/longitude`	Geographic coordinates
`pollutant_id`	Pollutant type (PM_{10}, $PM_{2.5}$, CO, NO_2, NH_3, O_3, SO_2)
`pollutant_min/max`	Min/max in reporting window
`pollutant_avg`	Mean reading (primary variable)

B. NFHS-5 Health and Socioeconomic Dataset (`health.xls`)

This dataset contains 111 rows and 136 columns representing state-level NFHS-5 indicator estimates. Each state has three rows corresponding to Urban, Rural, and Total (combined) population strata. Total-stratum rows were retained for state-level analysis. The dataset covers 37 states and union territories with socioeconomic, health, education, sanitation, and demographic indicators [7].

C. Dataset Integration

State names in the CPCB dataset use underscores as word separators (e.g., `Andhra_Pradesh`), while NFHS-5 uses spaces (`Andhra Pradesh`). After normalising both sources to lowercase, replacing underscores with spaces, and applying a targeted name-harmonisation dictionary for four residual mismatches ("NCT of Delhi" → "delhi"; "Maharastra" → "maharashtra"; "Andaman & Nicobar Islands" → "andaman and nicobar"; "Jammu & Kashmir" → "jammu and kashmir"), a total of **29 states and union territories** were successfully matched and merged. The eight unmatched NFHS territories (Goa, Manipur, Meghalaya, Ladakh, Lakshadweep, Dadra and Nagar Haveli & Daman and Diu, Andaman and Nicobar, India-aggregate) lack corresponding CPCB monitoring stations in the current snapshot.

III. METHODOLOGY

A. Data Preprocessing

AQI data. Rows with `pollutant_avg` equal to `NA` were dropped (138 rows; 4.0%). State names were normalised to lowercase with underscores replaced by spaces; a single-entry replacement maps `tamilnadu` (CPCB raw) to `tamil nadu` (standard). A composite state-level measure — the arithmetic mean of all `pollutant_avg` readings across pollutants, cities, and stations within each state — (`mean_pollutant_avg`) was computed to capture the overall pollution burden.

Health data. Only "Total" (combined urban+rural) rows were retained. Fourteen columns containing mixed text/numeric data were coerced to numeric type using `pd.to_numeric(..., errors='coerce')`. After name harmonisation, the merged dataset contained 29 state-level observations.

AQI categorisation. Each pollutant reading was classified using standard CPCB sub-index thresholds: Good (0–50), Satisfactory (51–100), Moderate (101–200), Poor (201–300), Very Poor (301–400), and Severe (>400).

B. Statistical Analysis

Pearson correlation coefficients and associated two-tailed p-values were computed between `mean_pollutant_avg` and ten selected socioeconomic/health indicators. Statistical significance was assessed at $\alpha = 0.05$ (critical $|r| \approx 0.37$ for $n = 29$).

C. K-Means Clustering

K-Means clustering was applied to four variables: `mean_pollutant_avg`, electricity access, improved sanitation, and female education — all standardised using z-score normalisation (`StandardScaler`). The optimal number of clusters was identified via the Elbow Method (evaluating inertia for $k = 2$ to 7). Clusters were visualised by projecting onto the pollution-vs.-female education axes.

D. Predictive Modelling

Eight socioeconomic features were used as predictors of `mean_pollutant_avg`: (1) female school attendance (%), (2) electricity access (%), (3) improved sanitation (%), (4) clean cooking fuel (%), (5) women's literacy (%), (6) women with ≥10 years schooling (%), (7) tobacco use — men (%), and (8) alcohol use — men (%).

Four regression models were evaluated: Linear Regression (OLS), Ridge Regression ($\alpha = 1.0$), Random Forest (200 trees, `max_depth` = 4), and Gradient Boosting (200 estimators, `max_depth` = 3).

All models were implemented as scikit-learn `Pipeline` objects [10] with `StandardScaler` as the first step, ensuring the scaler is re-fitted independently within each cross-validation fold and eliminating data leakage. Data were split 80/20 (train/test) with random seed 42. Five-fold cross-validation R^2 (CV R^2) was computed for each pipeline. Performance metrics include MAE, RMSE, test R^2, and CV R^2.

IV. RESULTS AND DISCUSSION

A. Pollutant Distribution

Table II summarises descriptive statistics for each pollutant.

Particulate matter constitutes the primary pollution concern. Mean PM_{10} (96.11 μg/m^3) exceeds the WHO annual guideline (15 μg/m^3) [9] by more than sixfold. Mean $PM_{2.5}$ (76.60 μg/m^3) exceeds the WHO guideline (5 μg/m^3) by over fifteenfold. All pollutants exhibit right-skewed distributions (means exceed medians), indicating hotspot stations driving the upper tail. Figure 1 illustrates these distributions.

TABLE II
DESCRIPTIVE STATISTICS OF POLLUTANT AVERAGE CONCENTRATIONS (μG/M^3 OR PPB)

Pollutant	n	Mean	Median	SD	Min	Max
PM_{10}	482	96.11	91.0	38.07	23.0	268.0
$PM_{2.5}$	486	76.60	65.0	41.30	2.0	243.0
CO	477	35.91	32.0	22.26	2.0	131.0
NO_2	481	29.22	24.0	21.64	1.0	138.0
O_3	459	21.95	18.0	15.87	1.0	143.0
SO_2	461	16.98	13.0	14.93	1.0	111.0
NH_3	427	6.16	5.0	4.67	1.0	44.0

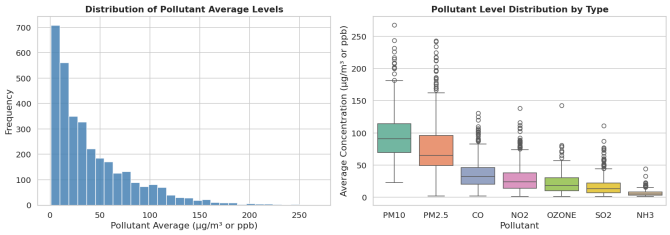

Fig. 1. Distribution of pollutant average levels: (left) frequency histogram across all readings; (right) box-plots by pollutant type.

B. AQI Category Distribution

Table III shows the breakdown of monitoring readings by CPCB AQI category.

While 69% of sub-index readings fall in the "Good" category — reflecting the inclusion of low-concentration pollutants (NH_3, SO_2) at baseline levels — the 9.7% at "Moderate" or worse represent significant public health exposure across large urban populations. It is important to note that these are individual pollutant sub-index readings, not composite AQI values; the composite AQI is defined as the maximum sub-index and would yield a higher share of elevated categories.

C. State-Level Pollution Rankings

Table IV presents the top-10 and bottom-5 states by mean composite pollution level.

Delhi's position at the top is consistent with established literature on the National Capital Region's severe air quality challenges [6]. Northern and eastern industrial states (Jharkhand, Haryana, Odisha) cluster in the high-pollution tier, while smaller north-eastern states and Himalayan union territories show consistently lower composite levels. Figure 2 provides a visual overview.

TABLE III
DISTRIBUTION OF MONITORING READINGS BY AQI CATEGORY ($n = 3{,}273$)

Category	Count	Share (%)
Good	2,268	69.3
Satisfactory	687	21.0
Moderate	299	9.1
Poor	19	0.6
Very Poor	0	0.0
Severe	0	0.0

TABLE IV
STATE-LEVEL MEAN COMPOSITE POLLUTION RANKINGS (μG/M^3 OR PPB)

Rank	State	Mean
1	Delhi	64.55
2	Himachal Pradesh	61.29
3	Tripura	53.00
4	Jharkhand	52.75
5	Haryana	48.24
6	Odisha	46.27
7	Gujarat	44.59
8	Punjab	43.74
9	Uttar Pradesh	43.58
10	Madhya Pradesh	43.13
25	Puducherry	22.00
26	Uttarakhand	20.67
27	Jammu & Kashmir	20.42
28	Nagaland	17.57
29	Sikkim	15.71

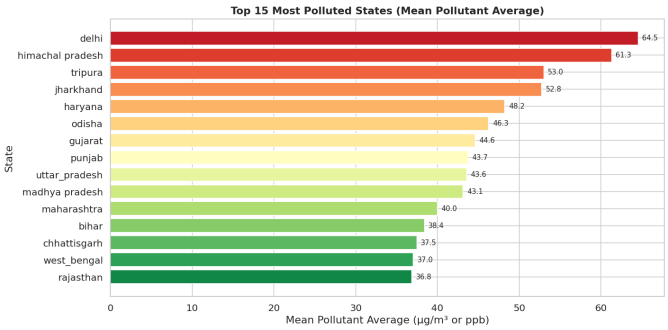

Fig. 2. Top 15 most polluted states by mean composite pollutant average (μg/m^3 or ppb).

D. Correlation Analysis

Table V reports Pearson r and two-tailed p-values for ten socioeconomic/health indicators ($n = 29$ states).

TABLE V
PEARSON CORRELATION WITH MEAN POLLUTANT AVERAGE ($n = 29$, CRITICAL $|r| \approx 0.37$ AT $\alpha = 0.05$)

Indicator	r	p	Sig.
Female Education (%)	-0.18	0.353	—
Electricity Access (%)	-0.13	0.519	—
Improved Sanitation (%)	-0.34	0.069	†
Clean Cooking Fuel (%)	-0.24	0.215	—
Women Literacy (%)	-0.18	0.350	—
Women Schooling $\geq$10yr (%)	-0.15	0.438	—
Tobacco Use — Men (%)	$+0.03$	0.859	—
Alcohol Use — Men (%)	-0.10	0.620	—
Infant Mortality (per 1,000)	$\mathbf{+0.45}$	**0.017**	$*$
Under-5 Mortality (per 1,000)	$\mathbf{+0.44}$	**0.021**	$*$

$*$ $p < 0.05$; † $p < 0.10$

Infant mortality ($r = +0.45$, $p = 0.017$) and under-five mortality ($r = +0.44$, $p = 0.021$) are the only indicators reaching conventional statistical significance. These positive associations suggest that states bearing higher composite pollution burdens also tend to have higher child mortality rates — consistent with the substantial literature linking particulate matter exposure to adverse perinatal and early-childhood health outcomes [5]. Improved sanitation shows a near-significant negative correlation ($r = -0.34$, $p = 0.069$).

Development indicators (female education, electricity access, clean cooking fuel) exhibit weak negative correlations, though none reach significance at $n = 29$. Three important caveats apply: (a) urbanisation acts as a confounder — Delhi ranks high in both infrastructure metrics and pollution, attenuating negative correlations; (b) the ecological fallacy applies — state-level aggregates mask within-state heterogeneity between urban hotspots and cleaner rural areas; (c) temporal mismatch exists between the CPCB snapshot (March 2026) and NFHS-5 data (2019–21). These results are therefore interpreted as exploratory associations rather than confirmed causal relationships. Figure 3 shows the full correlation matrix.

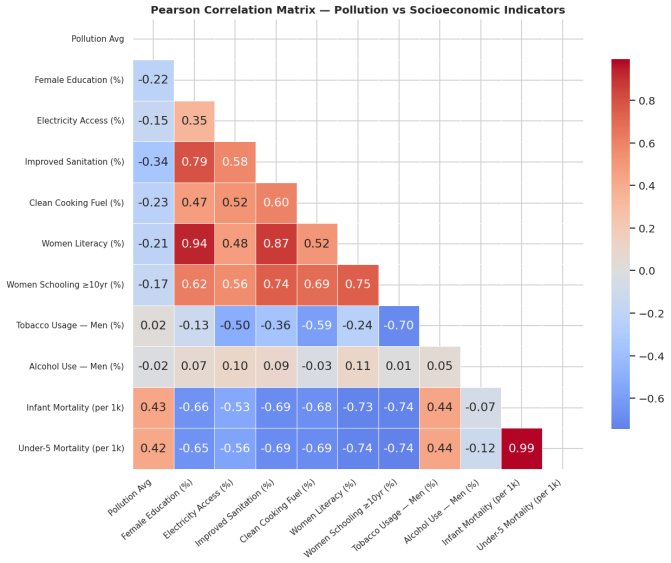

Fig. 3. Pearson correlation matrix — pollution vs socioeconomic indicators (lower triangle; colour scale: blue = negative, red = positive).

E. K-Means Clustering

The Elbow Method identified $k = 3$ as the optimal cluster number. Table VI summarises cluster profiles.

Cluster 0 groups high-pollution states with moderately developed infrastructure — consistent with the urbanisation-pollution nexus. Cluster 2 represents states combining lower pollution with high development indicators, broadly characterising southern and western India. Cluster 1 captures north-eastern and eastern states with moderate-to-lower development and lower pollution, largely reflecting lower industrial density. Figure 4 visualises the cluster assignments.

TABLE VI
K-MEANS CLUSTER PROFILES (MEAN VALUES, $k = 3$)

C	Poll.	Elec.	San.	FEd.	Representative States
0	∼47	91%	72%	79%	Delhi, Jharkhand, Haryana, Punjab, Odisha
1	∼28	90%	67%	74%	Assam, Bihar, Nagaland, Tripura
2	∼24	98%	93%	88%	Chandigarh, Gujarat, Karnataka, Kerala

Poll. = Mean Pollution; Elec. = Electricity; San. = Sanitation; FEd. = Female Education.

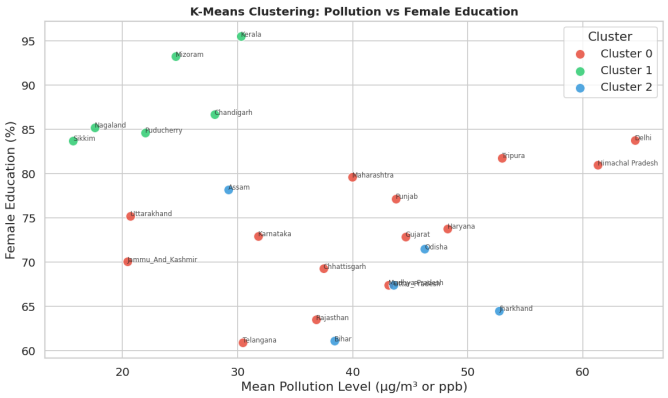

Fig. 4. K-Means clustering ($k = 3$) projected onto the Mean Pollution vs. Female Education axes. Each point represents a state; colours denote cluster membership.

F. Predictive Modelling

Table VII compares regression model performance on the held-out test set ($n_{\text{test}} = 6$) and via five-fold cross-validation.

TABLE VII
REGRESSION MODEL PERFORMANCE ($n = 29$, 80/20 SPLIT)

Model	MAE	RMSE	Test R^2	CV R^2
Linear Regression	16.24	18.81	−0.367	−1.998
Ridge Regression	15.79	18.00	−0.253	−1.618
Random Forest	**12.77**	**15.35**	**+0.090**	−1.515
Gradient Boosting	14.95	19.04	−0.402	−2.873

All models yield negative CV R^2 values, indicating no reliable generalisation beyond the training sample. Random Forest achieves a marginally positive held-out R^2 (+0.090), though this reflects the limited test fold size (six states). This outcome is a finding in itself: *socioeconomic features as measured here are insufficient to reliably predict state-level composite pollution at* $n = 29$. A larger longitudinal panel — incorporating temporal variation and within-state city-level observations — is necessary for meaningful predictive modelling.

Despite poor predictive accuracy, Random Forest feature importances (Figure 5) provide exploratory directional insight. Clean cooking fuel access and electricity access rank highest, followed by female education and women's literacy. This is consistent with the understanding that household energy

transition (from biomass to LPG/electricity) directly reduces particulate emissions, and that states with higher electrification tend to have greater environmental regulatory capacity.

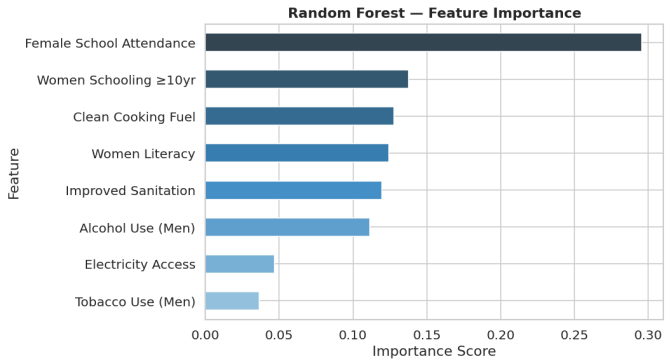

Fig. 5. Random Forest feature importance scores. Higher scores indicate greater influence on the predicted state-level mean pollution value.

V. Conclusion

This study presents an integrated analysis of CPCB real-time air quality data and NFHS-5 socioeconomic indicators across Indian states, contributing a reproducible, multi-method analytical framework. The principal findings are:

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1) **Particulate matter dominates India's air quality burden.** Mean PM_{10} and $PM_{2.5}$ concentrations exceed WHO guidelines by six- and fifteenfold respectively. Northern and eastern states bear a disproportionate pollution burden.
2) **Child mortality indicators are the only statistically significant correlates** of state-level pollution exposure (infant mortality: $r = +0.45$, $p = 0.017$; under-five mortality: $r = +0.44$, $p = 0.021$).
3) **K-Means clustering reveals three distinct state profiles:** a high-pollution/moderate-development cluster (northern industrial states), a lower-pollution/lower-development cluster (north-eastern states), and a lower-pollution/higher-development cluster (southern and western states).
4) **Predictive modelling is fundamentally constrained** by the small matched sample (29 states). No model achieved reliable held-out CV R^2.
5) **Clean cooking fuel and electricity access** are the most informative exploratory features, reinforcing the policy relevance of household energy transition programmes (Ujjwala Yojana, Saubhagya) as co-benefit strategies for air quality improvement.

A. Limitations

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- The merged dataset covers only 29 of 37 NFHS states.
- AQI data represents a single cross-sectional snapshot; causal inference requires longitudinal panel data.
- Temporal alignment between CPCB readings and NFHS-5 estimates (2019–21) is approximate.
- State-level aggregation masks substantial intra-state heterogeneity between urban and rural areas.
- No direct respiratory health outcome variables (e.g., hospital admissions) were available in the current datasets.

B. Future Work

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- Incorporate multi-year CPCB data (2017–present) to enable panel regression and trend analysis.
- Use city/district-level linkage between CPCB stations and NFHS Primary Sampling Units for higher-resolution analysis.
- Apply spatial regression models (Spatial Lag, Geographically Weighted Regression) to account for spatial autocorrelation.
- Integrate satellite-derived $PM_{2.5}$ data (NASA MERRA-2, MODIS) to extend coverage beyond ground-monitoring locations.

Acknowledgements

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Appendix

All analyses are fully reproducible using the accompanying Jupyter Notebook `CPCB_AQI_Analysis.ipynb` and the two raw data files `aqi.csv` and `health.xls`. Python dependencies are listed in `requirements.txt`. To reproduce:

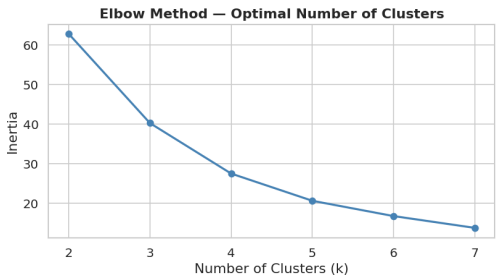

Fig. 9. Elbow Method for K-Means: inertia vs. number of clusters. The elbow at $k = 3$ justifies the chosen cluster count.

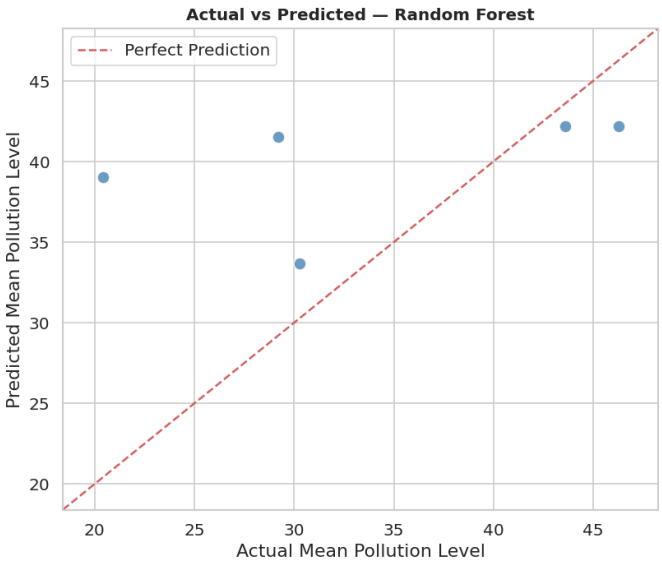

Fig. 10. Actual vs. predicted mean pollution level for the best model (Random Forest). The dashed red line represents perfect prediction.

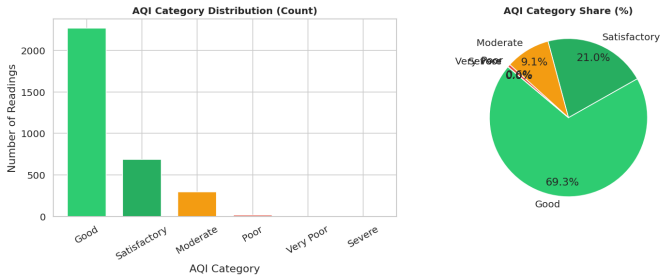

Fig. 6. AQI category distribution: (left) count bar chart; (right) percentage pie chart.

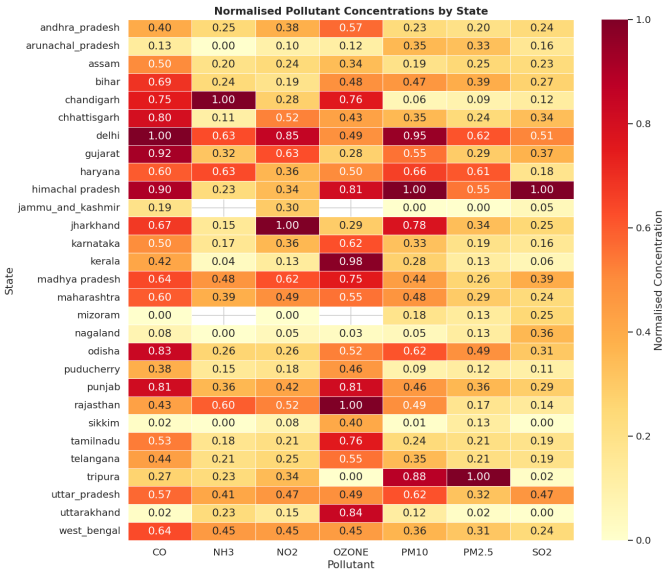

Fig. 7. Normalised pollutant concentrations by state (0–1 scale per pollutant column).

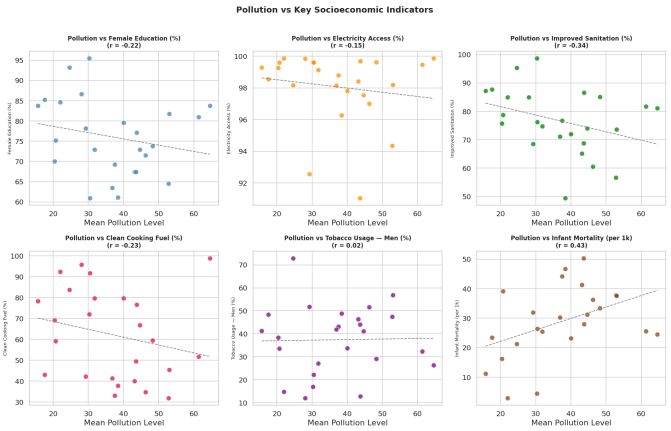

Fig. 8. Scatter plots — state-level mean pollution vs key socioeconomic indicators with OLS trend lines.

```
# 1. Install dependencies
pip install -r requirements.txt

# 2. Execute the notebook
jupyter nbconvert --to notebook \
  --execute CPCB_AQI_Analysis.ipynb \
  --output CPCB_AQI_Analysis_executed.ipynb
```

Listing 1. Reproducing the analysis

The notebook produces all 11 figures (Figures 1–11) automatically and prints all numerical results reported in this paper.

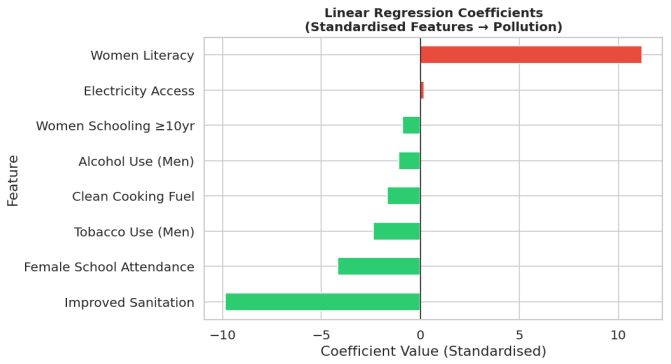

Fig. 11. Linear Regression standardised coefficients. Red bars indicate features positively associated with pollution; green bars indicate negative